

Model Selection via Validation and Cross-Validation

Variable Selection for Explanatory Linear Models

➤ Best Subset Selection

➤ Forward Selection

➤ Backward Selection

➤ Stepwise Selection

Note: If you're not interested in explanation/inference, regularized regression (LASSO/Ridge) is typically a better option for linear models.

Regularization has proven to provide better predictive models.

Variable Selection for Linear Models

- Validation dataset: Single holdout sample on which we compare candidate models.
- Cross-validation: Dividing our training data into k “folds” and using each fold as a validation set
- Input Variables: ALWAYS start by narrowing a list of reasonable input variables through exploratory analysis!

Using a Validation Set

...

When you have a LOT of data

Best Subset Selection

(Part 1)

- FOR $k=1, \dots, p$ where p is the number of variables in your dataset.
 - Consider all possible models with k variables
 - Build each model on your training data
 - Pick the best one to be your candidate model with k variables.

how do we define what model is best?

AIC

BIC

SBC

R^2

(On training data)

Best Subset Selection

(Part 2)

- Now you should have p different models from which to choose a final model.
- Run each model on the validation dataset and compare the MSE
- Pick the final model as the one with lowest MSE
- Train the final model on ALL data to update parameters.

Forward Selection

(Part 1)

- FOR $k=1, \dots, p$ where p is the number of variables in your dataset. Start with a null model (intercept only)
- Consider all models that add 1 additional variable to the current model.
- Choose the best among these models and update your current model accordingly

how do we define what model is best?

AIC

BIC

SBC

R^2

Forward Selection

(Part 2)

- Now you should have p different models from which to choose a final model.
- Run each model on the validation dataset and compare the MSE
- Pick the final model as the one with lowest MSE
- Train the final model on ALL data to update parameters.

Backward, Stepwise Selection

- Mimic the procedure for forward stepwise, only implement backward or stepwise selection when building candidate models.

Cross-Validation Approaches

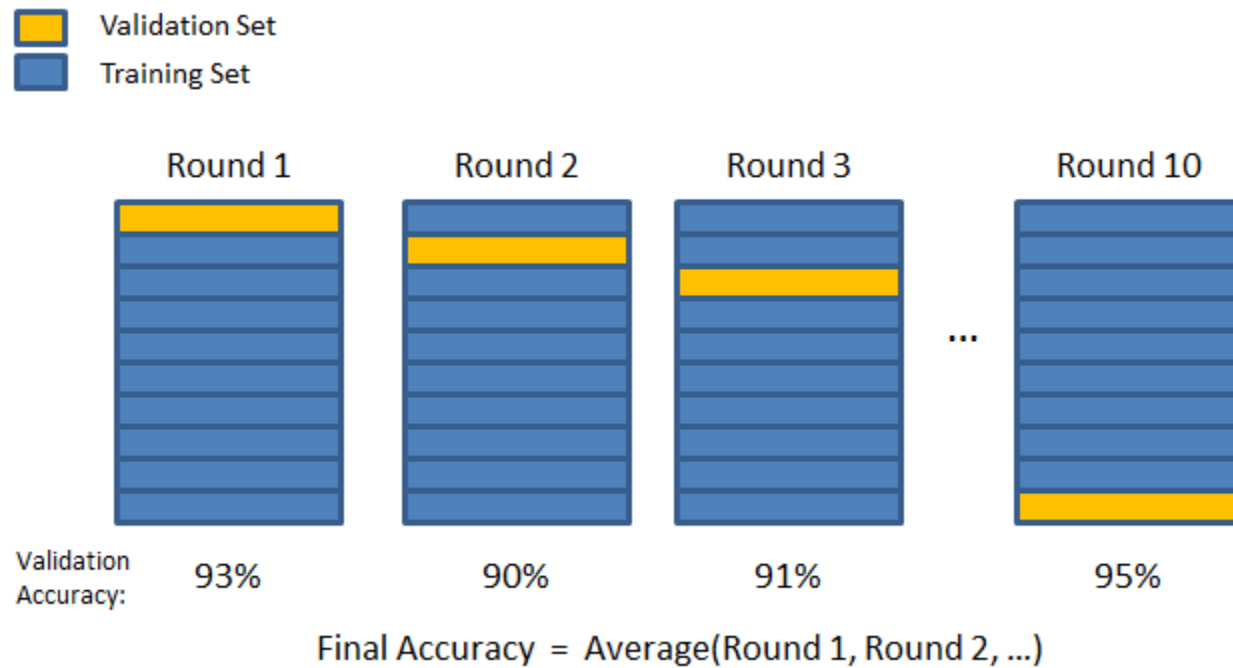
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Using CV Test error as model assessment

Vs.

'Tuning the parameter p '

Cross Validation



The Classic View

- Use CV to evaluate which model is 'best' at each step of a stepwise selection procedure.
- Once your CV error increases as you enter (forward/stepwise) or remove (backward) the next variable, you can stop.
- You've arrived at your final model. Update parameter estimates by training on all the data.

Forward Selection via Classical CV

- FOR $k=1, \dots, p$ where p is the number of variables in your dataset. Start with a null model (intercept only)
- Consider all models that add 1 additional variable to the current model.
- Choose the best among these models and update your current model accordingly

how do we define what model is best?

Average Error on CV

PROC GLMSELECT

```
proc glmselect data=bootcamp.ameshousing;  
class central_air house_style heating_qc  
masonry_veneer season_sold;  
model saleprice = basement_area age_sold  
garage_area lot_frontage Mas_Vnr_Area  
overall_qual overall_cond Mo_Sold central_air  
gr_liv_area lot_area_bedroom_abvgr  
fireplaces full_bathroom  
half_bathroom deck_porch_area  
central_air house_style  
heating_qc masonry_veneer season_sold  
/selection=stepwise(stop=CV) cvmethod=random(10);  
run;
```



Tells SAS to stop once the CV error sums of squares increases for all possible variable additions



Tells SAS to randomly partition the data into 10 folds for CV.

The SAS System

The GLMSELECT Procedure

Forward Selection Summary

Step	Effect Entered	Number Effects In	NumberParms In	SBC	CV PRESS
0	Intercept	1	1	6325.9189	4.33598E11
1	Overall_Qual	2	2	6098.9909	1.98552E11
2	Basement_Area	3	3	5978.0627	1.31304E11
3	Garage_Area	4	4	5921.0264	1.07616E11
4	Age_Sold	5	5	5884.3149	9.4219E10
5	Gr_Liv_Area	6	6	5827.6080	7.67161E10
6	Overall_Cond	7	7	5800.9482	6.94417E10
7	Fireplaces	8	8	5775.9024	6.33433E10
8	Lot_Area	9	9	5761.3840	6.00916E10
9	Bedroom_AbvGr	10	10	5751.8058*	5.78982E10*

* Optimal Value of Criterion

Selection stopped at a local minimum of the cross validation PRESS.

Stop Details

Candidate For	Effect	Candidate CV PRESS	Compare	CV PRESS
Entry	Central_Air	5.81109E10	>	5.78982E10

PRESS= Predicted Residual Error Sums of Squares

Of all the variables, Overall_Qual provided the best 1-variable model according to predictive accuracy on CV

After controlling for variables 1-9, the 10 variable model that performed best is that with central_air. However, it performed worse than the 9 variable model.

PROC GLMSELECT

Parameter Estimates
output result from
training the chosen model
on ALL the training data

Parameter Estimates				
Parameter	DF	Estimate	Standard Error	t Value
Intercept	1	4425.084079	6026.424745	0.73
Basement_Area	1	24.521045	2.728084	8.99
Age_Sold	1	-499.829967	38.871809	-12.86
Garage_Area	1	29.017984	5.270182	5.51
Overall_Qual	1	7358.134341	969.774133	7.59
Overall_Cond	1	5607.668408	753.377769	7.44
Gr_Liv_Area	1	43.980117	5.100262	8.62
Lot_Area	1	1.413736	0.267444	5.29
Fireplaces	1	7835.174483	1475.007345	5.31
Bedroom_AbvGr	1	-5354.827064	1375.509534	-3.89

Another View: Tuning the parameter p

- Cross validation is used to determine the optimal number of variables for the model.
- In essence, the variable selection technique becomes part of your model flow.

Least Squares Regression
with the variables Age,
Income, and Credit_Score

Least Squares Regression
with forward selection
choosing 3 variables.

Old way to think about
specifying your model

New way to think about
specifying your model

Best Subset Selection

(Part 1)

- For each CV fold:
 - Consider all possible models with k variables
 - Build each model on your training data (all but 1 fold)
 - Pick the best one (by performance on the validation fold) to be your candidate model with k variables.
- At the end of that procedure, you should have p models for each fold.

Best Subset Selection

(Part 2)

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Avg CV Error
1 variable	Model(1,1)	Model(1,2)	Model(1,3)	Model(1,4)	Model(1,5)	97
2 variables	Model(2,1)	Model(2,2)	Model(2,3)	Model(2,4)	Model(2,5)	81
3 variables	Model(3,1)	Model(3,2)	Model(3,3)	Model(3,4)	Model(3,5)	63
4 variables	Model(4,1)	Model(4,2)	Model(4,3)	Model(4,4)	Model(4,5)	75
5 variables	Model(5,1)	Model(5,2)	Model(5,3)	Model(5,4)	Model(5,5)	79

- Conclude after this process that best subset selection with 3 variables works best.
- You will then use best subset selection with 3 variables on the entire dataset.
- You will not restrict yourself to any specific variables.

R Lab

➤ For additional practice, check out:

Introduction to Statistical Learning Section 6.5 Lab 1